



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to MAHA FUEL AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

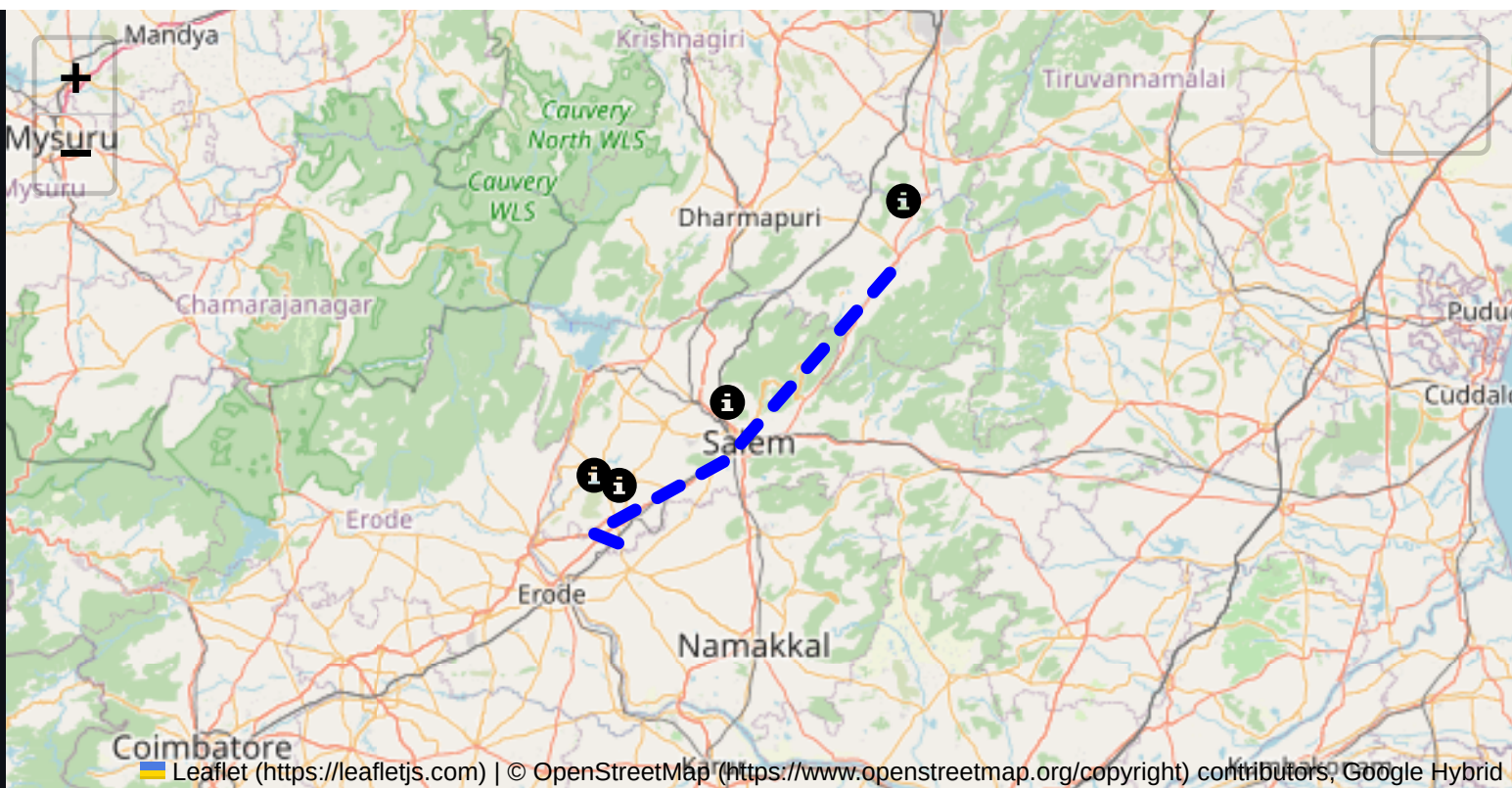
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

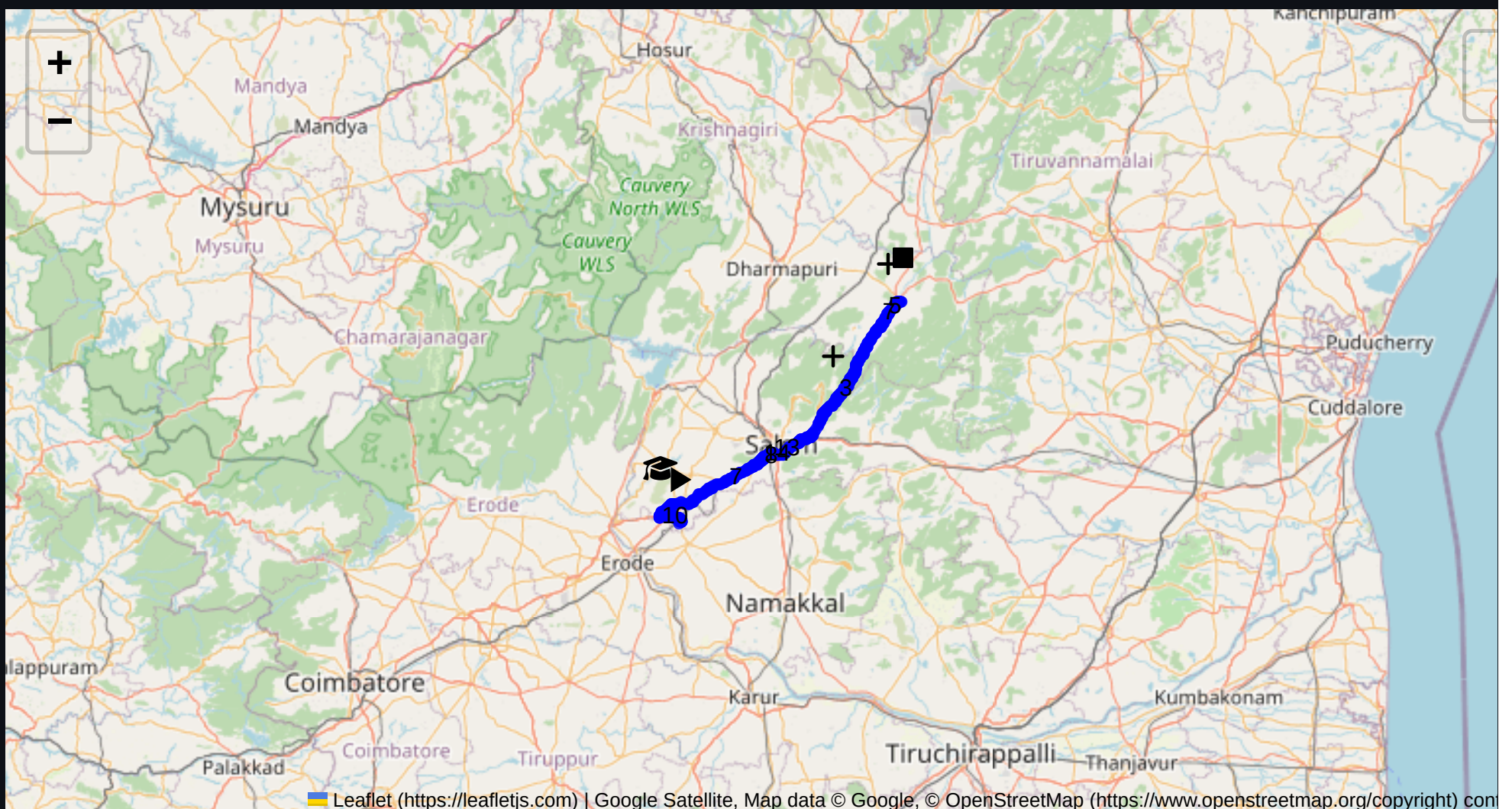
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 117.10 km
Estimated Duration: 2.8 hours
Adjusted Duration (Heavy Vehicle): 3.5 hours
Start: (11.4381, 77.8734)
End: (12.04408088, 78.49145516)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to 2FVR+5R5, Oldpet, Harur, via FR59+7CQ, Padaiveedu, and Coimbatore Rd, Salem, covers approximately 117.10 kilometers. It traverses through the towns and smaller regions of Tamil Nadu, offering a varied landscape of urban and semi-urban areas. The journey

primarily follows regional highways and some stretches of national highways, ensuring a relatively direct and accessible passage.

Typical Weather Conditions and Potential Weather-Related Hazards

- **Seasonal Weather:** Tamil Nadu generally experiences tropical weather with distinct wet (monsoon) and dry seasons.
- **Monsoon Hazards:** The region is prone to heavy rainfall during the monsoon (June to September), which can lead to waterlogging and muddy conditions, particularly in rural sections.
- **Summer Heat:** Intense heat during the summer months might affect vehicle performance, especially for heavy vehicles.

Traffic Patterns and Congestion-Prone Areas

- **Peak Hours:** Typically, peak traffic times are between 8:00 AM to 10:00 AM and 5:00 PM to 8:00 PM in urban areas.
- **Congestion:** Salem and the entrances to smaller towns can experience congestion and slow-moving traffic, particularly near markets or industrial areas.

Road Quality and Infrastructure

- **Highway Conditions:** The highways are generally well-maintained but can feature sections with potholes and rough surfaces, especially after monsoon periods.
- **Rural Roads:** Less-traveled rural roads may be narrower and more difficult to navigate for large trucks, with potential obstructions like small bridges or roadside stalls.

Alternative Routes for Emergencies

- Detours through smaller local roads can be utilized in emergency situations, though they might lengthen travel time.
- Always check road conditions and construction updates that might provide faster or safer options when emergencies arise.

Local Regulations Affecting Hazardous Material Transport

- **Permits & Passes:** Truck drivers must ensure they hold the correct permits, particularly when transporting hazardous materials, following state and national guidelines.
- **Transport Timing Restrictions:** Movement of hazardous materials might be restricted during peak times or in highly populated areas to minimize risk.

Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- **Historical data** does not indicate frequent major incidents, but intersections in busy areas have seen minor collisions involving large vehicles. Hazardous material spillage is a critical concern in accident zones.

Environmental Considerations and Sensitive Areas

- **Ecological Zones:** Passing near agricultural lands requires care to prevent contamination from spillage.
- **Wildlife Areas:** Be cautious in areas proximal to wildlife crossings, especially where signposted.

Communication Coverage

- **Mobile Network:** Generally good but can be patchy in isolated rural stretches. Drivers should ensure communications systems are capable of handling these gaps.

Estimated Emergency Response Times

- **Urban Areas (Salem):** Faster response time, typically within 15-30 minutes.
- **Rural Areas:** Could exceed 30-45 minutes depending on road conditions and distance from service centers.

Overall Risk Assessment Summary

The route generally provides a viable transit path, with manageable risks primarily associated with weather and road conditions. Key risk factors include potential congestion in urban areas and adverse weather affecting road traction. Traffic and infrastructure in smaller towns require careful navigation by large vehicles. Ensuring up-to-date information on permits, local road advisories, and pre-planned communication strategies can mitigate these risks effectively. Training drivers in emergency response and spill containment protocols is essential to maintain safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	High	11.45353, 77.85697	15 KM/Hr
5	Turn	Medium	11.45853, 77.85139	30 KM/Hr
6	Turn	Medium	11.45906, 77.85132	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit	
7	Turn	High	11.46307, 77.84941	15 KM/Hr	
8	Turn	High	11.45542, 77.81725	15 KM/Hr	
9	Turn	High	11.45636, 77.81678	15 KM/Hr	
10	Turn	Medium	11.53578, 77.95802	30 KM/Hr	
11	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr	
12	Turn	High	11.62106, 78.10767	15 KM/Hr	
13	Blind Spot	Blind Spot	11.61217, 78.10439	10 KM/Hr	
14	Turn	Medium	11.62537, 78.12040	30 KM/Hr	
15	Turn	Medium	11.63283, 78.12757	30 KM/Hr	
16	Turn	Medium	11.65815, 78.20461	30 KM/Hr	
17	Turn	Medium	11.66359, 78.20647	30 KM/Hr	
18	Turn	Medium	11.67128, 78.22971	30 KM/Hr	
19	Turn	High	11.67131, 78.23138	15 KM/Hr	
20	Turn	High	11.67177, 78.23140	15 KM/Hr	
21	Turn	Medium	11.67416, 78.23864	30 KM/Hr	
22	Turn	Medium	11.68005, 78.24245	30 KM/Hr	
23	Turn	Medium	11.79058, 78.32176	30 KM/Hr	
24	Turn	Medium	11.80467, 78.33345	30 KM/Hr	
25	Turn	Medium	11.83500, 78.35302	30 KM/Hr	
26	Turn	Medium	11.99586, 78.44568	30 KM/Hr	
27	Turn	Medium	12.04278, 78.48346	30 KM/Hr	
28	Turn	High	12.04668, 78.48397	15 KM/Hr	
29	Turn	Medium	12.04645, 78.48495	30 KM/Hr	
30	Turn	Medium	12.04651, 78.48546	30 KM/Hr	

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium

		type	name	coordinates	speed_limit	risk_level
	4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
	5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
	6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
	8	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
	13	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
	14	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
	16	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
	17	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
	18	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
	19	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
	20	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
	21	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
	22	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
	23	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
	24	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
	25	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
	26	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
	28	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
	30	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
	33	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
9	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
27	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
29	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
31	school	Indian Matriculation Higher secondary school	12.006299, 78.4531204	30 km/h	Medium
32	school	Indian Matriculation Higher Secondary school	12.0061755, 78.4533388	30 km/h	Medium
34	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
35	school	Chinnan Kuppam Government High School	12.0190729, 78.4581955	30 km/h	Medium
36	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
37	school	Jayam Vidhyalaya Matriculation Higher Secondary School	12.0264103, 78.4656553	30 km/h	Medium
38	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45906, 77.85132



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45636, 77.81678



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.53578, 77.95802



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.62145, 78.10740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.62106, 78.10767



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61217, 78.10439



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62537, 78.12040



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63283, 78.12757



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65815, 78.20461



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66359, 78.20647



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67128, 78.22971



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67131, 78.23138



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67177, 78.23140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67416, 78.23864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68005, 78.24245



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.79058, 78.32176



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.80467, 78.33345



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83500, 78.35302



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99586, 78.44568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04278, 78.48346



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.04668, 78.48397



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.04645, 78.48495



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.04651, 78.48546
